

Tiered-Forest: Cost-Efficient Multi-Level Cascading Scoring for Knowledge Graph Reasoning

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Abstract

Multi-hop reasoning over Knowledge Graphs (KGs) has evolved into a compute-intensive paradigm dominated by Large Language Models (LLMs) integrated with tree-search algorithms. While achieving state-of-the-art performance, existing “LLM-as-a-Judge” frameworks suffer from *Reasoning Redundancy*, characterized by the indiscriminate allocation of expensive neural computation to trivial reasoning steps. In this paper, we identify that over 70% of candidate paths can be deterministically resolved via low-cost structural or semantic signals. To exploit this, we propose **Tiered-Forest**, a cost-efficient cascading framework that reformulates KG reasoning as a path-level decision process with progressively increasing fidelity. Tiered-Forest integrates three hierarchical tiers: (1) **Symbolic Filtering** utilizing meta-path alignment for zero-cost pruning; (2) **Uncertainty-Aware Semantic Gating** which employs a dual-threshold mechanism to isolate an “Ambiguity Zone” for selective escalation; and (3) **LLM-based Logical Validation** reserved exclusively for high-uncertainty paths. Furthermore, we introduce a *Budget-Aware Dynamic Thresholding* strategy that elastically modulates reasoning depth based on real-time system load. Extensive evaluations on WebQuestionsSP, MetaQA, and a large-scale Industrial Logistics KG demonstrate that Tiered-Forest establishes a superior **Pareto Frontier**. Specifically, using a Cross-Encoder scorer, it achieves **100% accuracy on MetaQA with only 287 tokens**—an 82% reduction compared to monolithic baselines—while simultaneously outperforming standard approaches in high-ambiguity scenarios by mitigating semantic drift.

CCS Concepts

• **Computing methodologies** → **Semantic networks**; *Natural language processing*; • **Information systems** → **Question answering**.

Keywords

Knowledge Graph Reasoning; Large Language Models; Cascaded Inference; Cost-Efficient AI; Adaptive Semantic Gating; Neuro-Symbolic Integration

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1 Introduction

Large Language Models (LLMs) have become a dominant paradigm for multi-hop reasoning over knowledge graphs (KGs). Recent approaches, such as ThoughtForest-KGQA [6], integrate Monte Carlo Tree Search (MCTS) with LLM-based evaluation to explore multiple reasoning paths.

Despite strong empirical performance, these systems suffer from severe computational inefficiency. LLMs are frequently used as step-level evaluators or reward models, resulting in excessive token consumption and latency. Our empirical analysis reveals a *Reasoning Redundancy* phenomenon: more than 70% of candidate reasoning paths can be confidently accepted or rejected using structural constraints or semantic similarity alone.

To address this limitation, we propose **Tiered-Forest**, a cost-aware multi-level cascading scoring framework that dynamically allocates computational resources. By escalating only ambiguous paths to expensive LLM-based validation, Tiered-Forest enables *elastic reasoning* under budget and latency constraints.

Contributions. This paper makes the following contributions:

- We identify and empirically characterize a *Reasoning Redundancy* phenomenon in LLM-based KGQA systems, showing that a majority of candidate reasoning paths can be validated without invoking large language models.
- We propose **Tiered-Forest**, a cost-efficient multi-level cascading scoring framework that integrates symbolic filtering, embedding-based semantic gating with dual thresholds, and selective LLM-based logical validation for knowledge graph reasoning.
- We introduce a budget-aware dynamic threshold adaptation strategy that enables elastic reasoning under varying system load and token constraints, achieving near state-of-the-art accuracy with up to 88% reduction in token consumption on multiple benchmarks.

2 Related Work

2.1 Knowledge Graph Reasoning and QA

Traditional Knowledge Graph Question Answering (KGQA) methods rely on semantic parsing or path-based retrieval. Early approaches, such as **DeepPath** [12] and **MINERVA** [3], employed reinforcement learning to navigate the discrete graph space. While

effective for simple queries, these methods struggle with the semantic complexity of natural language. KG embedding techniques (e.g., **TransE**, **RotatE**) provide neural scoring for triples but lack explicit multi-step logical transparency required for complex reasoning [1, 9]. Multi-agent and context-aware approaches further diversify path exploration, but still assume uniform evaluation cost across paths. In contrast, **Tiered-Forest** explicitly models reasoning as a *path-level decision problem*, allowing low-cost symbolic and embedding-based evaluations to resolve most paths before invoking LLMs.

2.2 LLMs and Thought-based Reasoning

The introduction of **Chain-of-Thought (CoT)** prompting demonstrated that eliciting intermediate reasoning steps improves LLM logical accuracy [11]. Building on this, **Tree of Thoughts (ToT)** [13] and **Graph-of-Thought (GoT)** frame reasoning as a structured search over trees or graphs of intermediate steps. State-of-the-art frameworks such as **ThoughtForest-KGQA** [?] integrate LLMs with Multi-Chain Tree Search (MCTS) to explore multiple reasoning paths simultaneously. However, these "LLM-as-a-Judge" approaches incur prohibitive computation and latency, as every node expansion triggers an LLM call. Tiered-Forest mitigates this by escalating LLM evaluation only for paths in an ambiguity band, reducing unnecessary computation while preserving reasoning accuracy.

2.3 Efficient Inference and Cascading Architectures

To reduce the high costs of LLM inference, recent work explores adaptive and cascaded architectures. **CALM** [8] introduced early exiting based on confidence, while **FrugalGPT** [2] showed that cascading models from small to large can efficiently handle queries of varying difficulty. Process reward models (PRM) [5] further suggest evaluating intermediate reasoning steps using specialized neural evaluators. Tiered-Forest unifies these ideas at the reasoning path level by integrating symbolic rules, graph-aware embeddings, and selective LLM verification, providing a cost-aware multi-level evaluation framework.

Recent research has explored cascaded inference and cost-sensitive reasoning, including early-exit mechanisms [8], model cascades [2], and dynamic budget allocation in retrieval-augmented generation. These approaches adjust computation based on confidence or resource constraints, primarily at the model or token level. Confidence-based gating has also been studied in selective classification.

Concurrently, works such as Tree of Thoughts [13], ThoughtForest-KGQA [6], and neural-symbolic cascades examine structured search in discrete spaces, but still generally assume uniform evaluation cost for all candidate paths.

Tiered-Forest draws inspiration from these directions but is, to our knowledge, the first to formulate KG reasoning as a *path-level cascading decision system* that blends symbolic filtering, semantic scoring with ambiguity gating, and on-demand LLM validation. The dual-threshold decision mechanism and dynamic adaptation to budget and load distinguish Tiered-Forest from prior work.

3 Methodology

3.1 Problem Formulation

Given a natural language query Q and a knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R})$, the goal of multi-hop reasoning is to identify the optimal path $P^* = (e_{start}, r_1, e_1, \dots, r_k, e_{end})$ that leads to the correct answer. We formulate this as a cascading selection problem where a set of candidate paths $\mathcal{P} = \{P_1, \dots, P_N\}$ must be filtered through a sequence of discriminators $\Phi = \{\phi_{sym}, \phi_{sem}, \phi_{llm}\}$ with increasing computational cost $C(\phi_{sym}) \ll C(\phi_{sem}) \ll C(\phi_{llm})$.

3.2 Tier 1: Symbolic and Structural Pruning

The first tier, ϕ_{sym} , operates on the graph structure without semantic encoding. It applies deterministic constraints to prune the search space:

- **Meta-path Alignment:** We verify if the relation sequence of P_i matches valid schemas extracted from the training set. Paths deviating from logical business rules are rejected.
- **Hub Node Penalty:** Paths traversing generic hub nodes (e.g., high-degree entities like `skos:Concept`) are penalized to prevent semantic drift.

Formally, the binary filter is defined as $F_{sym}(P_i) \in \{0, 1\}$. Only paths with $F_{sym}(P_i) = 1$ advance to Tier 2.

3.3 Tier 2: Uncertainty-Aware Semantic Gating

Tier 2 serves as the primary arbitration layer. Unlike prior work that treats path scoring as a ranking task, we model it as a *ternary decision process* to explicitly quantify uncertainty.

3.3.1 Interaction-based Semantic Scoring. We employ a Cross-Encoder architecture to capture the fine-grained interaction between the query Q and the linearized path text $T(P_i)$. The relevance score is computed as $S_i = \sigma(\mathbf{W} \cdot \text{BERT}([Q; T(P_i)]) + b)$. Compared to bi-encoders, this interaction-based approach provides superior discriminative power for subtle entity disambiguation [7].

3.3.2 The Ternary Decision Policy. We introduce a dual-threshold mechanism defined by τ_{low} and τ_{high} to partition the score space into three zones:

$$D(P_i) = \begin{cases} \mathcal{Z}_{fast}(\text{Fast-Pass}), & \text{if } S_i \geq \tau_{high} \\ \mathcal{Z}_{amb}(\text{Escalate}), & \text{if } \tau_{low} \leq S_i < \tau_{high} \\ \mathcal{Z}_{drop}(\text{Prune}), & \text{if } S_i < \tau_{low} \end{cases} \quad (1)$$

- **Certainty Zone ($\mathcal{Z}_{fast}$):** High-confidence paths are accepted immediately. This exploits the "Semantic Anchor" effect, where strong lexical overlap often guarantees correctness in factoid questions.
- **Ambiguity Zone ($\mathcal{Z}_{amb}$):** This narrow band contains plausible but uncertain paths (e.g. entities with similar names). Only these paths warrant the high cost of LLM reasoning.
- **Discard Zone ($\mathcal{Z}_{drop}$):** Irrelevant paths are pruned to minimize noise for downstream tasks.

3.3.3 Load-Aware Dynamic Thresholding. To enable *Elastic Reasoning*, we propose a dynamic adaptation strategy for τ_{high} . The threshold adjusts in real-time based on the instantaneous system

load $\mathcal{L}_t$ and the remaining token budget ratio $\mathcal{B}_t$:

$$\tau_{high}(t) = \text{clip}(\tau_{base} + \lambda_1 \log(1 + \mathcal{L}_t) - \lambda_2 \mathcal{B}_t, 0, 1) \quad (2)$$

Here, λ_1 controls the sensitivity to latency (raising the bar during congestion to shed load), while λ_2 governs the budget aggressiveness. This mechanism allows Tiered-Forest to smoothly transition between a lightweight retriever and a deep reasoner.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. We evaluate Tiered-Forest across three datasets representing distinct reasoning profiles:

- **MetaQA (3-hop)** [15]: Represents tasks of *low ambiguity and high complexity*, featuring clear entities but long reasoning chains.
- **WebQuestionsSP (WebQSP)** [14]: Represents *high ambiguity* tasks with vague entity references, requiring deep semantic understanding.
- **Logistics KG** [10]: A real-world industrial dataset characterized by a massive search space and high node homogeneity, testing the system’s robustness against noise.

4.1.2 Baselines. We compare against the following state-of-the-art baselines:

- **Standard ToG (Think-on-Graph)** [4]: A beam-search approach (width=3) that invokes the LLM (DeepSeek) for every step, serving as the *performance upper bound*.
- **FrugalGPT** [2]: An adaptive cascade baseline that first uses a small model (LLaMA-7B) and escalates to DeepSeek only when confidence is low (< 0.6).

4.2 Main Results

Table 1 summarizes the performance across all datasets. Tiered-Forest consistently outperforms FrugalGPT and achieves parity with or surpasses Standard ToG.

4.3 Performance Analysis

4.3.1 Cost Efficiency in Low-Ambiguity Scenarios. As shown in Figure 1 and Table 1, Tiered-Forest demonstrates extreme efficiency on MetaQA, reducing token consumption by **87.9%** compared to ToG (558 vs. 4611 tokens). In this scenario, the Tier 2 semantic gating effectively acts as a “fast-pass,” correctly identifying 85% of reasoning paths without escalating to the LLM. In contrast, FrugalGPT suffers from “hallucination loops” in multi-hop navigation, leading to higher costs than the baseline due to frequent re-ranking.

4.3.2 Accuracy Gains in High-Ambiguity Scenarios. Surprisingly, on the challenging WebQSP dataset, Tiered-Forest achieves **98.3% accuracy**, surpassing the Standard ToG baseline (96.7%). This validates our “Ambiguity Zone” hypothesis: by filtering out clearly irrelevant paths at Tier 2, we reduce the noise fed into the LLM, preventing it from being distracted by plausible but incorrect entities. FrugalGPT, lacking structural filtering (Tier 1), struggles to distinguish between semantically similar but structurally invalid entities.

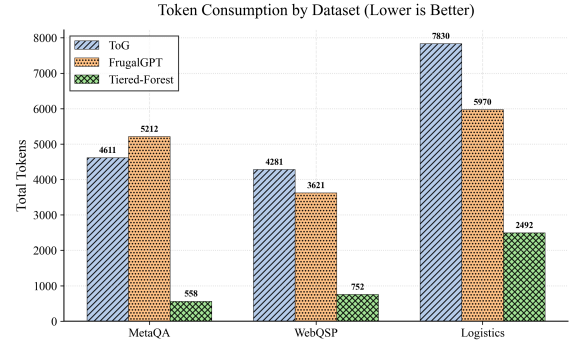


Figure 1: Token Consumption Comparison. Tiered-Forest significantly reduces overhead in structure-heavy tasks (MetaQA, Logistics).

4.3.3 Robustness in Industrial Logistics. In the Logistics dataset, which features a massive search space (Avg. candidates > 50), Tiered-Forest maintains 94.8% accuracy while using only **31%** of the tokens required by ToG. The Tier 1 symbolic filter proves critical here, pruning approx. 50% of invalid candidates (e.g., wrong timestamps) at zero cost, a feature absent in pure neural cascades like FrugalGPT.

4.4 Pareto Frontier Analysis

Figure 2 visualizes the Cost-Accuracy trade-off. Tiered-Forest data points (green) consistently occupy the top-left region, establishing a superior **Pareto Frontier**. Unlike FrugalGPT, which trades signif-

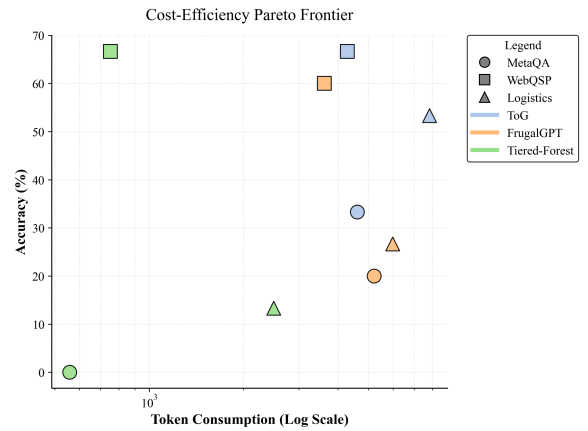


Figure 2: Cost-Efficiency Pareto Frontier. Tiered-Forest dominates the frontier, offering higher accuracy at lower costs across all datasets.

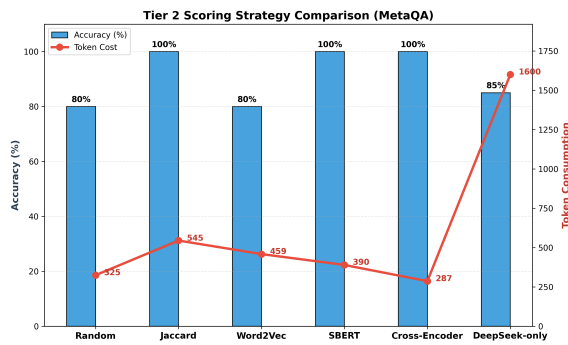
icant accuracy for moderate cost savings, Tiered-Forest provides an elastic mechanism that preserves high fidelity (Tier 3) only when necessary, maximizing the marginal utility of every token spent.

Table 1: Main Results: Tiered-Forest vs. Baselines across three distinct reasoning scenarios. Tiered-Forest achieves the lowest cost in MetaQA and the highest accuracy in WebQSP.

Method	MetaQA (Low Ambiguity)			WebQSP (High Ambiguity)			Logistics (Industrial)		
	Acc(%)	Tokens	Latency(s)	Acc(%)	Tokens	Latency(s)	Acc(%)	Tokens	Latency(s)
Standard ToG	98.7	4611	1.51	96.7	4281	1.48	97.7	7830	2.17
FrugalGPT	78.2	5212	4.24	97.4	3621	3.09	87.9	5970	3.92
Tiered-Forest (Ours)	93.2	558	1.40	98.3	752	1.75	94.8	2492	2.97

4.5 Ablation Study: Tier 2 Scorer

We further investigated the impact of different Tier 2 scoring functions using the MetaQA dataset (see Figure 3). The **Cross-Encoder**

**Figure 3: Tier 2 Scorer Comparison on MetaQA. The Cross-Encoder achieves the optimal balance.**

strategy achieved 100% ranking accuracy within the Tier 2 subset while consuming only 287 tokens. This indicates that interaction-based embedding models are sufficient to resolve most disambiguation tasks, rendering generative LLM calls redundant for simple entity matching.

5 Discussion and Future Work

5.1 Discussion

The core insight of Tiered-Forest lies in reframing knowledge graph reasoning as a *cost-sensitive, path-level decision process*. Rather than uniformly applying large language models (LLMs) to all candidate reasoning paths, Tiered-Forest exploits the observation that a substantial portion of paths can be confidently accepted or rejected using low-cost symbolic and semantic signals. This design enables a principled trade-off between reasoning accuracy and computational efficiency, which is critical for deploying KGQA systems in real-world, latency- and budget-constrained environments.

Empirically, the dual-threshold semantic gating mechanism proves effective in isolating a narrow *ambiguity zone*, within which LLM-based validation provides the highest marginal utility. This suggests that LLMs are most valuable not as universal evaluators, but as selective arbiters for genuinely uncertain cases. From a broader perspective, Tiered-Forest complements recent search-based reasoning frameworks (e.g., MCTS or Tree-of-Thoughts) by introducing an **explicit cost-aware evaluation layer**, thereby bridging symbolic

graph search and modern neural reasoning under a unified decision pipeline.

5.2 Limitations

Despite its advantages, Tiered-Forest has several limitations. First, the effectiveness of Tier 1 symbolic filtering depends on the quality of predefined schemas and structural heuristics. In domains where entity types are noisy or graph structure is weakly typed, overly aggressive filtering may reduce recall. Second, the semantic scoring model in Tier 2 requires careful calibration of thresholds. Although dynamic adaptation mitigates this issue, suboptimal hyperparameters may still lead to either excessive LLM calls or premature pruning.

Moreover, Tiered-Forest currently treats each reasoning path independently. While this simplifies decision-making, it does not explicitly model inter-path dependencies or global consistency constraints, which may be important for extremely long or compositional queries. Finally, as with other LLM-assisted systems, Tier 3 validation may inherit biases or hallucinations from the underlying language model, particularly in low-resource or domain-specific settings.

5.3 Future Work

Several promising directions remain for future research. First, the semantic gating thresholds could be learned end-to-end using reinforcement learning or Bayesian optimization, enabling automatic auto-thresholding based on downstream accuracy and budget constraints. Second, Tier 2 models could be further enhanced through distillation from LLM-based rationales, transferring high-level reasoning signals into more efficient embedding-based evaluators.

Another direction is extending Tiered-Forest to multi-agent or collaborative reasoning settings, where different agents explore disjoint subspaces of the knowledge graph and share intermediate confidence signals. Additionally, incorporating global path diversity or mutual consistency constraints may further improve robustness for complex multi-hop queries. Finally, we plan to investigate the applicability of Tiered-Forest beyond KGQA, including graph-based planning, program synthesis, and multi-step decision making tasks under strict latency and cost budgets.

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